

RAPHAEL ESQUIVEL

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EDUCATION

Harvey Mudd College, Claremont, CA

Aug 2025 - May 2029

B.S., Mathematics & Physics

GPA: 3.88/4.0

Relevant coursework: Electromagnetic Fields, General Relativity, Quantum Field Theory, Nanomaterials, Engineering Systems, Abstract Algebra I, Intermediate Probability, Numerical Analysis, Readings in Algebraic Geometry

Arizona State University, Phoenix, AZ

Aug 2024 - Apr 2025

Non-Degree Seeking Student

GPA: 4.0/4.0

Relevant coursework: Mathematical Methods Physics II, Introductory Modern Physics, Quantum Physics II, Classical Particles, Fields, and Matter II

EXPERIENCE

Repair Steward, HMC Makerspace

Aug 2026 - Present

Student Researcher, Logarithmic Gromov-Witten Theory (HMC)

Nov 2025 - Present

- Analysis of the tropical and combinatorial structure of relative stable maps to $(\mathbb{P}^n, H)$ and their boundary behavior in moduli spaces.

Student Researcher, Quantum Quasars Lab (HMC)

Sep 2025 - Present

- Designing and assembling a free-space quantum communication system for a kilometer-scale demonstration of quantum position verification.
- Developed of an ultra-low-latency RF synchronization link and FPGA-based control for RF communication and photonics.
- Built and aligned optical experimental setup and fabricated custom optomechanical components.

AWARDS

Karl Menger Memorial Award (2nd), American Mathematical Society

May 2025

Awarded \$1000 at Regeneron ISEF for project *Lie Groups and Algebroids in Lagrangian Reduction*.

SKILLS

Programming languages

Python, Javascript, C/C++, R, Mathematica, Julia, Verilog

Software

COMSOL, Xilinx Vivado, Git, Jupyter

Lab equipment

Free-space optics, lasers, SPAD, Pockels cells, optical fiber

Manufacturing

FDM, SLS, lathe, mill, 3D printer maintenance

Languages

English (native), French (native), Spanish (proficient)

RESEARCH INTERESTS

Algebraic geometry, quantum field theory, Gromov-Witten theory